

Reference cohort composition confounds distributional evaluation of medical image generators: a depth-resolved audit of radiology foundation-model features

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Abstract

Distributional metrics such as the Fréchet Inception Distance compare a set of generated images against a finite set of real ones. In natural-image benchmarks the real reference is a collection of independent photographs, so its composition is rarely examined. Medical imaging violates that assumption twice over: adjacent slices from one patient are near-duplicates, and multi-site cohorts carry systematic acquisition differences. We show that these two properties, not generator quality, can dominate the reported number. On BraTS brain MRI, the maximum mean discrepancy (MMD^2) between two *real* patient cohorts drawn from the same dataset reaches 0.096, exceeding the 0.080 measured between a real cohort and the output of an unconditional DDPM; a reference built by taking the first 500 filenames in sorted order — a plausible but reproducibility-poor convenience — selects a 17-subject block that is systematically atypical of the cohort, and inflates the reported score by $1.9\times$ relative to a subject-diverse reference of identical size. We propose replacing the conventional permutation test, which asks whether real and generated samples share a distribution and is therefore rejected in every configuration we tested, with an *empirical inter-cohort reference distribution* built from distances between disjoint real patient blocks; this reports whether a generator differs from the reference by more than routine inter-cohort variation, and it makes the dependence of any reported score on cohort size explicit. Using this protocol we then audit which encoder depth

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of a frozen radiology vision transformer (RadioDINO-s16) supports which evaluation task, at all twelve blocks. Depth is strongly task-dependent and the final block is not the best choice for any task we measure: distributional discriminability peaks at a mid-to-late plateau (L7–L9) and is 22% lower at L12; near-duplicate detection improves monotonically with depth; degradation monotonicity fails at L1–L2; and k -nearest-neighbour precision/recall coverage fails its own mode-drop validation at *every* depth, with recall rising rather than falling as diversity is removed. For per-image novelty scoring, the choice of scoring rule matters more than depth: a shrinkage Mahalanobis distance reaches AUC 0.972 where distance to the real centroid reaches 0.846. We release the evaluation code and report a concrete protocol: report the number of distinct subjects behind every reference set, calibrate against a cohort null, audit read-out depth rather than defaulting to the final block, and do not report k -NN coverage in this feature space without validating it.

Keywords: generative models, medical image synthesis, evaluation metrics, maximum mean discrepancy, dataset bias, foundation models

1. Introduction

Generative models are increasingly proposed as a response to data scarcity in medical imaging, where privacy constraints, annotation cost and disease rarity limit dataset size (Kebaili et al., 2023; Kazerouni et al., 2023). Whether a synthetic dataset is fit for a downstream purpose depends on being able to measure how far it departs from real data, and the measurement instruments in common use were designed for a different setting. The Fréchet Inception Distance (FID) (Heusel et al., 2017) and Kernel Inception Distance (KID) (Bińkowski et al., 2018) embed images with an ImageNet classifier and summarise the gap between two feature distributions as a scalar. Both have been transplanted into medical image synthesis largely unchanged.

Most methodological criticism of this transplant has focused on the *embedding*: an ImageNet backbone encodes texture statistics that need not correspond to radiological structure, motivating replacements built on domain-pretrained encoders (Dohmen et al., 2025; Woodland et al., 2024). That criticism is well founded, and we take a domain-pretrained encoder as our starting point. But it leaves a second assumption unexamined. Every distributional metric is a comparison against a *finite real reference set*, and its value is a property of that reference as much as of the generator. In natural-image

benchmarks the reference is tens of thousands of independent photographs, so its composition is close to the population it stands for. Medical references violate this in two specific ways:

1. **Slices are not independent samples.** A reference of N axial slices may come from N patients or from a dozen; neighbouring slices of one volume are near-duplicates, so the effective sample size is closer to the number of subjects than to N .
2. **Cohorts are heterogeneous.** Public collections such as BraTS aggregate several contributing institutions with different scanners and protocols. Which subjects enter the reference therefore carries a site signature.

This paper asks how much of a reported metric value those two properties account for, and finds that on BraTS they can account for more than the generator does. Two real patient cohorts drawn from the same dataset can differ by more, in the metric, than a real cohort differs from unconditional DDPM output. A reference assembled by the most convenient available route — the first N filenames in sorted order — selects a block that is systematically atypical of the cohort and inflates the reported score by nearly a factor of two relative to a subject-diverse reference of the same size. Neither effect is visible in the number itself; both are invisible in a paper that reports N but not the number of subjects behind it.

This observation has a constructive consequence. The significance machinery usually attached to kernel two-sample tests asks whether the real and generated samples were drawn from the same distribution. For realistic generative models, exact equality of the generated and real distributions is not the hypothesis of practical interest, and in every configuration we tested the test rejected it at overwhelming significance regardless of generator quality; we measure permutation z -scores above 400 for every generator we consider, including one that is a different imaging modality. The comparison a practitioner needs is different: *is this generator further from the real reference than two real patient cohorts are from each other?* That question has a natural empirical null, obtained by measuring distances between disjoint real subject blocks. We formalise this inter-cohort reference distribution, show that it orders generators the permutation test cannot, and show that it makes the cohort-size dependence of any reported score explicit.

With a defensible reference protocol in place, we then address a question that the shift to domain-pretrained encoders raises but has not settled: a transformer has many layers, and papers using such encoders as metric back-

bones generally read the last one. We audit all twelve blocks of RadioDINO-s16 (Pérez-García et al.) against four evaluation tasks — distributional fidelity, near-duplicate (memorization) detection, mode-drop coverage, and per-image novelty scoring — and find that depth is strongly task-dependent and that the final block is the best choice for none of them. We also find that one widely used diagnostic, k -NN precision/recall (Kynkäänniemi et al., 2019), fails its own validation at every depth, in a direction explained by neighbourhood-radius inflation.

Contributions.

1. We quantify reference-cohort composition as a confound in distributional evaluation of medical generative models, and show that on BraTS real-versus-real cohort distance exceeds real-versus-generated distance (Section 4.1).
2. We propose an empirical inter-cohort reference distribution that complements the permutation test with a comparison against inter-cohort heterogeneity, and report how it scales with cohort size (Section 4.2).
3. We provide a depth-resolved audit of a radiology foundation model across four evaluation tasks, showing that the a-priori final-layer choice is sub-optimal for all of them (Section 4.4).
4. We report that k -NN recall fails its mode-drop sanity check at every evaluated depth in this feature space, with a mechanism, and that per-image scoring is more sensitive to the choice of distance than to depth (Sections 4.5 and 4.6).
5. We audit ten published medical generation papers and find that none reports the number of distinct subjects behind its real reference set, which is what makes the confound invisible in practice (Section 4.3).
6. We release code implementing the protocol and all experiments.

2. Related work

Distributional metrics and their estimators. FID models both feature sets as Gaussians and reports the Fréchet distance between them (Heusel et al., 2017). The estimator is biased at finite sample size, and the bias depends on N in a way that makes values from different studies incomparable (Chong and Forsyth, 2020). KID (Bińkowski et al., 2018) replaces the Gaussian

assumption with an unbiased kernel MMD² estimator (Gretton et al., 2012), which removes the sample-size bias in the estimator but not the dependence on what the reference set contains. CMMD (Jayasumana et al., 2024) keeps the MMD estimator and substitutes CLIP features for Inception features. Our estimator is a standard unbiased multi-bandwidth RBF MMD²; the contribution here is not the estimator but what it is calibrated against.

Domain-specific embeddings. Several groups have argued that ImageNet features are inappropriate for medical evaluation and proposed radiology-pretrained substitutes, including RadImageNet-based FID variants (Dohmen et al., 2025) and self-supervised chest and brain encoders (Pérez-García et al.). Woodland et al. (2024) compare feature extractors for medical FID and report that the choice changes conclusions. The Fréchet Radiomic Distance (Konz et al., 2026) moves in a different direction, replacing learned features with handcrafted radiomic descriptors. This literature establishes that embedding choice matters; it does not examine the reference set, which is what we address.

Beyond a single scalar. Precision/recall decompositions (Sajjadi et al., 2018; Kynkäänniemi et al., 2019) and α -precision/ β -recall (Alaa et al., 2022) separate sample quality from coverage, and memorization audits ask whether a generator reproduces training examples (Carlini et al., 2023; Akbar et al., 2023). These diagnostics are typically computed in the same embedding as the fidelity score and at a single, unexamined depth. We evaluate them across depth and find that one of them does not transfer to this feature space at all.

Dependence on the reference set. The observation that a distributional score depends on what it is measured against is not new in the natural-image literature, and we do not claim it as ours. Chong and Forsyth (2020) showed that the FID estimator is biased in the reference sample size, and proposed extrapolating to an infinite-sample value. Most directly, Lee and Pak (2026) study the *geometry* of the reference dataset, showing across six datasets that distributional density and effective rank explain much of how FID responds as sample quality improves, and that concentrated references produce more favourable FID trends than dispersed ones. Their recommendation — report distributional metrics together with descriptors of the reference set — is one we endorse and extend.

Our question is different in a way that matters for medical imaging, and it is worth stating precisely which axis is new here. That literature treats the reference as a set of independent images and asks how its geometry in feature space affects the metric. We ask what happens when the reference is *not* a set of independent images: when it is a set of slices drawn from a much smaller number of patients, so that its effective sample size is governed by the subject count rather than the image count, and when which subjects are drawn carries systematic acquisition structure. That distinction has three consequences with no counterpart in the natural-image setting — a real-versus-real comparison that can exceed the real-versus-generated one, a null that shrinks with subject diversity while the reported score barely moves, and a subject-level construction protocol — and it is those we contribute. The sampling-frame question of whether the reference cohort represents the intended deployment population is a fourth issue, which neither that literature nor this paper addresses.

Dataset bias and site effects. That multi-site medical datasets carry scanner and protocol signatures strong enough to be recovered by a classifier is well documented (Glocker et al., 2019; Wachinger et al., 2021), as is the resulting need for subject-level rather than image-level data splits (Varoquaux and Cheplygina, 2022). To our knowledge this literature has not been connected to generative evaluation metrics, where the reference set plays the role that a test split plays in discriminative work, and where the same leakage and composition concerns therefore apply.

3. Materials and methods

3.1. Data

Real reference (BraTS). We use axial slices extracted from the BraTS 2021 training collection (Baid et al., 2021), giving 38,781 slices from 1,251 subjects (median 31 slices per subject) stored as 256×256 8-bit PNG. Slices are extracted from the FLAIR volume, intensity-normalised per volume to the $[0, 1]$ range after clipping at the 1st and 99th percentiles, and retained where the brain mask covers at least 5% of the frame. Subject identity is recoverable from the filename, which is what makes the cohort analysis in this paper possible; we note that a reference distributed as a flat directory of anonymised slice images would not permit it.

Generated and shifted sets. Four comparison sets of 500 images each are used throughout: (i) **DDPM**, an unconditional denoising diffusion probabilistic model (Ho et al., 2020) with a UNet backbone initialised from `google/ddpm-celebahq-256` and fine-tuned on the BraTS slices for 50 epochs at 256×256 (batch 16, learning rate 10^{-5} , fp16), sampled with the standard 1000-step reverse process; (ii) **WDM-3D (MRI)**, axial slices from a 3D wavelet diffusion model (Friedrich et al., 2024) trained on BraTS, a same-modality generator of visibly lower slice-level quality; (iii) **LIDC CT**, axial slices from the same 3D architecture trained on LIDC-IDRI chest CT (Armato III et al., 2011), a cross-modality shift; and (iv) **Retinal**, colour fundus images from RetinaMNIST (Yang et al., 2023), a far-domain shift. Sets (ii)–(iv) provide an a-priori severity ordering against which a metric can be checked.

Lesion-level evaluation (LGG). For the lesion-specificity analysis we use the lower-grade glioma collection of Buda et al. (2019), which supplies expert tumour masks registered to the slices, allowing perturbation to be confined to annotated tumour tissue or to area-matched healthy control regions.

3.2. Reference set construction

Let $\mathcal{S}$ be the set of subjects and $\mathcal{I}_s$ the slices of subject s . A reference of N slices is specified by a subject block $B \subseteq \mathcal{S}$ and a draw from $\bigcup_{s \in B} \mathcal{I}_s$. We distinguish three protocols:

HEAD the first N filenames in sorted order. This is the protocol a directory listing yields by default. Because BraTS filenames are ordered by subject identifier, it selects the lowest-numbered subjects and, at $N = 500$, spans only 17 of them.

BALANCED(m) m subjects sampled uniformly at random, with slices drawn round-robin — one per subject before any second — and shuffled within each subject. This fixes N while varying subject diversity, and avoids drawing only low-index slices, which are volume-edge positions and anatomically atypical.

DISJOINT two BALANCED blocks with no shared subject, used wherever a real-versus-real comparison is required.

Unless stated otherwise, results use BALANCED(100) for the reference and DISJOINT blocks for real-versus-real comparisons. Subject-level disjointness matters: a real-versus-real null built by splitting slices *within* subjects

measures near-duplicate distance, not cohort distance, and is optimistically small.

3.3. Feature extraction

We embed images with RadioDINO-s16 (Pérez-García et al.), a ViT-S/16 pretrained on radiological images by self-distillation, held frozen throughout. Images are resized to 224×224 , replicated to three channels, and normalised with the encoder’s training statistics. For each of the $L = 12$ transformer blocks we take the CLS token of that block’s output, giving twelve 384-dimensional representations per image from a single forward pass. Reading every depth costs no more than reading one, which is what makes the audit in Section 4.4 inexpensive.

We write $\phi_\ell(x) \in \mathbb{R}^{384}$ for the block- ℓ embedding. Features are L_2 -normalised before any distance computation.

3.4. Distributional discrepancy

For real features $X = \{\phi_\ell(x_i)\}_{i=1}^m$ and generated features $Y = \{\phi_\ell(y_j)\}_{j=1}^n$ we use the unbiased MMD² estimator (Gretton et al., 2012)

$$\widehat{\text{MMD}}^2(X, Y) = \frac{1}{m(m-1)} \sum_{i \neq i'} k(x_i, x_{i'}) + \frac{1}{n(n-1)} \sum_{j \neq j'} k(y_j, y_{j'}) - \frac{2}{mn} \sum_{i,j} k(x_i, y_j), \quad (1)$$

with a multi-bandwidth Gaussian kernel $k(u, v) = \frac{1}{5} \sum_c \exp(-\|u-v\|^2 / (2c \sigma_{\text{med}}^2))$, $c \in \{\frac{1}{4}, \frac{1}{2}, 1, 2, 4\}$, where σ_{med}^2 is the median pairwise squared distance on the pooled sample. Averaging over bandwidths avoids committing to a single kernel scale; we report the single-bandwidth comparison in Section 4.4. The estimator is unbiased for a fixed kernel and a fixed pair of sampling distributions. That is a weaker guarantee than it is often read as: it says nothing about whether the empirical reference represents the population it stands for. We therefore hold the image count fixed in every cohort-composition experiment, so that what varies is which subjects constitute the reference and not how many images it contains.

3.5. An empirical inter-cohort reference distribution

The conventional null for Eq. (1) is $H_0^{\text{perm}} : P_{\text{real}} = P_{\text{gen}}$, tested by pooling the two samples, relabelling at random, and recomputing. We report it for comparison, but it is not the hypothesis of interest: no generator satisfies it, and rejecting it says nothing about whether a generator is usable.

We therefore define a null from the data itself. Partition $\mathcal{S}$ into K disjoint blocks $B_1, \dots, B_K$ of m subjects each, and draw N slices from each by the BALANCED protocol. The *cohort-heterogeneity null at cohort size m* is the set of pairwise distances

$$\mathcal{N}_m = \{ \widehat{\text{MMD}}^2(R_a, R_b) : 1 \leq a < b \leq K \}, \quad (2)$$

where R_a is the draw from block B_a . For a generated set G scored against reference R_1 , we report the cohort-calibrated statistic

$$z_{\text{cohort}} = \frac{\widehat{\text{MMD}}^2(R_1, G) - \text{mean } \mathcal{N}_m}{\text{sd } \mathcal{N}_m}, \quad p_{\text{cohort}} = \frac{1 + |\{v \in \mathcal{N}_m : v \geq \widehat{\text{MMD}}^2(R_1, G)\}|}{1 + |\mathcal{N}_m|}. \quad (3)$$

$\mathcal{N}_m$ depends on m , so z_{cohort} is reported together with the cohort size that produced it. This is a feature rather than a nuisance: it states in the metric itself how much resolution a reference of that size affords.

On confidence intervals. Under H_0^{perm} the statistic in Eq. (1) is a degenerate U -statistic, and the ordinary nonparametric bootstrap is not consistent for it (Arcones and Giné, 1992); an m -out-of- n bootstrap or subsampling is required. We therefore never place a bootstrap interval on a null comparison. Every interval reported here is computed under the alternative — real against generated, where the statistic is non-degenerate and the ordinary bootstrap is consistent — using 500 resamples of each sample independently. Null behaviour is characterised by the empirical distribution $\mathcal{N}_m$ instead, which requires no asymptotic argument.

3.6. Diagnostics under audit

Each diagnostic below is computed at every block $\ell \in \{1, \dots, 12\}$ from the same cached features, and scored by a figure of merit that does not presume which depth should win.

Distributional fidelity. Figure of merit: the permutation z -score of $\widehat{\text{MMD}}^2(R, G)$ against a label-shuffling null, which measures how many null standard deviations separate the observed discrepancy from chance. We additionally check whether each depth recovers the a-priori severity ordering $\text{real} < \text{DDPM} < \text{WDM-3D} < \text{CT}$.

Memorization. A known fraction r of the generated set is replaced by copies of real images perturbed with Gaussian jitter of scale σ . The detector flags a generated image when its nearest-neighbour distance to the reference falls below the median within-reference nearest-neighbour distance. Figure of merit: the mean absolute error between the flagged rate and r , over $r \in \{0, 0.1, 0.2, 0.3, 0.4\}$. Sweeping σ matters, because at very small σ the task saturates and every depth succeeds.

Coverage. The generated set is truncated to simulate mode drop, retaining a fraction $1 - d$ for $d \in \{0, 0.2, 0.4, 0.6, 0.8\}$, and k -NN precision/recall (Kynkäänniemi et al., 2019) is computed with $k = 5$. Figure of merit: the Spearman correlation between d and recall, which must be strongly *negative* for the diagnostic to be usable — removing diversity must reduce measured coverage.

Per-image novelty scoring. Four rules score a single image against the reference: distance to the reference centroid; 1-NN and 5-NN distance; and a Mahalanobis distance using a covariance estimate shrunk toward a scaled identity ($\hat{\Sigma}_\varepsilon = (1 - \varepsilon)\hat{\Sigma} + \varepsilon\bar{\lambda}I$, $\varepsilon = 0.1$), which is necessary because a 384-dimensional covariance estimated from 500 samples is ill-conditioned. Figure of merit: ROC-AUC separating held-out real images from a positive class. We report two positive classes of unequal difficulty — cross-modality CT and the DDPM’s own output — because a task that saturates cannot distinguish rules or depths.

Degradation monotonicity. Gaussian noise of increasing σ is applied to the generated set. Figure of merit: Spearman correlation between σ and $\widehat{\text{MMD}}^2$, which should be +1.

3.7. Baselines and implementation

FID and KID are computed with `torchmetrics` 1.9.0 at its default 2048-dimensional InceptionV3 pool3 features, from `uint8` inputs at 256×256 , letting the library perform its own resize and normalisation. We flag a pitfall we encountered: a second, hand-rolled FID built on `torchvision`’s ImageNet-weight InceptionV3 with ImageNet normalisation produces values on a different scale for the same images, and mixing the two within one study yields tables that disagree by tens of FID points while appearing to describe the same comparison. All FID and KID numbers in this paper come from

one implementation. CMMD (Jayasumana et al., 2024) uses CLIP ViT-L/14 features with the authors’ unbiased Gaussian MMD estimator.

Experiments run on a single NVIDIA RTX 4070 Laptop GPU. Extracting all twelve depths for 500 images takes 2.5 s; one MMD^2 evaluation at $N = 500$ takes 4.5 s including feature extraction. Building a cohort null from 24 blocks requires 24 extractions and 276 MMD evaluations. All experiments use seed 42 unless a seed sweep is reported.

4. Results

4.1. Reference cohort composition changes the reported score

The default protocol selects an atypical block. Taking the first 500 filenames in sorted order from the BraTS slice directory yields images from 17 subjects. Randomly sampling 500 slices from the same directory spans 413. Both satisfy a caption reading “ $N = 500$ real slices”, and they give different answers: MMD^2 at L12 against the same DDPM samples is 0.150 for the HEAD reference and 0.077 for the random one, a factor of 1.95.

Subject count alone does not explain this. Table 1 compares 17-subject blocks drawn from different positions in the cohort, each supplying 500 slices drawn identically. Every block other than the first gives $\widehat{\text{MMD}}^2 \approx 0.082$; the first gives 0.152. The effect is therefore specific to *which* subjects the protocol selects, not to how many, and it is not a sample-size artefact.

We can state what we measured but not, from this data, its cause. BraTS 2021 does not release a per-subject site label, so we cannot confirm that the lowest-numbered subjects share a contributing institution; acquisition-site or protocol grouping correlated with subject-identifier order is the most economical explanation for a block-structured effect of this size, but it remains a hypothesis. What matters for the argument here is weaker and established directly: subject identifiers are not exchangeable, so any protocol that selects subjects by identifier order is selecting a structured subset of the cohort.

Real-versus-real distance can exceed real-versus-generated distance. Table 2 places that site effect on the same scale as the quantity the metric is meant to report. Two arbitrary 17-subject real cohorts differ by $\widehat{\text{MMD}}^2 = 0.017$. The first-17 block differs from the rest of the real cohort by 0.096. A subject-diverse real reference differs from DDPM output by 0.080. The distance between two sets of *real* BraTS images therefore exceeds the distance between real images and generated ones, whenever one of them is the atypical block

Table 1: Seventeen-subject reference blocks drawn from different positions in the BraTS subject list, each supplying 500 slices, scored against the same DDPM sample. Only the block that the default protocol selects is anomalous.

Reference block (17 subjects, $N = 500$)	$\widehat{\text{MMD}}^2$ at L12
First 17 subjects (what HEAD selects)	0.152
Middle 17 subjects	0.084
Last 17 subjects	0.082
Random 17 subjects (seed 0)	0.081
Random 17 subjects (seed 2)	0.082

Table 2: Distances on a common scale (L12, $N = 500$, all cohorts 17 subjects so that cohort size is not a confound). The third row involves no generated images at all, yet exceeds the second.

Comparison	Content	$\widehat{\text{MMD}}^2$
Real cohort A vs. real cohort B	real vs. real	0.017
Real cohort A vs. DDPM	real vs. generated	0.080
First-17 block vs. real cohort A	real vs. real	0.096
First-17 block vs. DDPM	real vs. generated	0.152

that the default reference protocol happens to select. We stress the scope of this claim: it is not that typical real cohort pairs exceed the generated distance — they do not, at 0.017 — but that the worst-case pair, which is the one the convenient protocol selects, does.

Cohort diversity mostly moves the null, not the signal. Holding $N = 500$ fixed and sweeping the number of contributing subjects (Table 3, Fig. 1) separates two effects. The real-versus-generated distance is comparatively stable, rising by 9% at L12 between a 400-subject and a 17-subject reference. The real-versus-real null is not: it rises from 0.0000 to 0.0072 over the same range. FID moves more than either MMD variant, from 67.9 to 85.3 (1.26 $\times$), consistent with its estimator depending on how well the reference covariance is estimated. The practical consequence is that reducing cohort diversity mainly destroys the evaluator’s ability to distinguish a real cohort from a generated one, rather than changing the headline number by much — a failure a single reported scalar cannot expose.

Table 3: Reference cohort diversity at fixed $N = 500$ slices, mean $\pm$ s.d. over three draws. Subject diversity moves the real-versus-real null far more than it moves the real-versus-generated score.

Subjects	$\widehat{\text{MMD}}^2$ L8	$\widehat{\text{MMD}}^2$ L12	null L8	null L12	FID
17	0.1021 ± 0.0046	0.0801 ± 0.0030	0.0082	0.0072	85.3 ± 1.8
25	0.1026 ± 0.0037	0.0799 ± 0.0025	0.0088	0.0074	82.8 ± 1.6
50	0.0975 ± 0.0035	0.0751 ± 0.0026	0.0024	0.0024	71.7 ± 1.0
100	0.0958 ± 0.0016	0.0739 ± 0.0005	0.0011	0.0010	72.5 ± 0.4
200	0.0966 ± 0.0032	0.0749 ± 0.0023	0.0003	0.0002	68.4 ± 0.6
400	0.0957 ± 0.0013	0.0736 ± 0.0008	0.0000	0.0000	67.9 ± 0.4
400 $\rightarrow$ 17	$\times 1.07$	$\times 1.09$	+0.0082	+0.0072	$\times 1.26$

4.2. An empirical inter-cohort reference distribution

Table 4 reports the null of Eq. (2) at three cohort sizes, together with each generator’s position within it. Two observations follow.

First, the permutation test does not discriminate here. It assigns $z > 340$ and $\hat{p}$ at the resolution floor to every generator at every cohort size, including generators as different from one another as a fine-tuned in-domain DDPM and colour fundus photography. It answers a question whose answer was never in doubt.

Second, and more revealing, the two nulls move in *opposite* directions as the reference improves. Going from a 17- to a 100-subject reference, the inter-cohort statistic for the DDPM rises from $z_{\text{cohort}} = 15.7$ to 89.0. The standardisation should not be over-read: the numerator is almost unchanged across cohort sizes (0.0776, 0.0752, 0.0766), and the rise is driven by the null spread contracting $5.4\times$. What a more subject-diverse reference buys is resolution, not a larger effect. Over the same range the permutation statistic *falls*, from $z = 423$ to $z = 347$. The permutation null is built by shuffling labels within the pooled sample, so a more heterogeneous real set inflates the null variance and depresses the statistic: a better reference makes the conventional test report less significance. The inter-cohort reference distribution does not have this defect, because it is constructed from real cohorts only.

For an interpretable effect size we report the generator distance as a multiple of the *mean* real-versus-real cohort distance. With a 17-subject reference the DDPM sits at $5.5\times$ the mean real cohort pair; at 50 subjects,

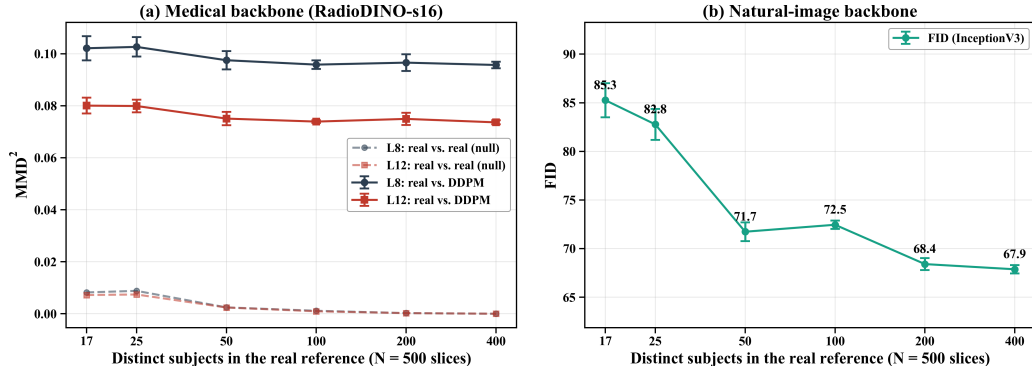


Figure 1: Reference cohort diversity at fixed $N = 500$. (a) RadioDINO-s16 MMD^2 against DDPM output (solid) and against a held-out real probe (dashed). The generated distance is comparatively flat, whereas the real-versus-real null grows sharply as diversity falls. (b) FID on the same references.

14.5 $\times$; at 100 subjects, 30.3 $\times$.

We use the mean rather than the maximum deliberately. A ratio to the largest observed cohort distance is a ratio to an order statistic, whose expectation grows with the number of pairs compared, so it is not comparable across conditions that contribute different numbers of pairs. Our 17- and 50-subject conditions both use $K = 24$ blocks and therefore exactly 276 pairs, and the effect is already established between them with the pair count held fixed: the ratio to the mean rises from 5.5 $\times$ to 14.5 $\times$, and the ratio to the maximum from 2.3 $\times$ to 6.9 $\times$. The 100-subject condition admits only $K = 12$ disjoint blocks at this cohort size and so rests on 66 pairs; its maximum is drawn from a smaller sample and we do not compare it to the others on that basis. Table 4 reports the mean, the 95th percentile and the maximum so the reader can apply whichever summary they prefer.

4.3. What do published papers actually report?

Sections 4.1–4.2 show that reference composition changes the reported number. A fair objection is that we constructed a deliberately poor selection rule and then demonstrated that it is poor. Whether the finding matters depends on an empirical question about the field: do published papers report enough about their reference set for a reader to know which rule was used?

We audited ten papers that evaluate medical image generation, spanning brain MRI, chest X-ray, cardiac MRI and CT, from 2022 to 2025, and including two papers whose subject is evaluation methodology rather than

Table 4: Inter-cohort reference distribution at L12 and generator positions within it. z_{cohort} rises with a better reference; the permutation z falls. “ $\times$ mean” and “ $\times$ max” are the generator distance as a multiple of the mean and the largest real-versus-real cohort distance; the former is preferred because the latter is an order statistic whose expectation depends on the number of pairs. The 17- and 50-subject rows rest on the same $K = 24$ blocks and 276 pairs and are directly comparable; the 100-subject rows rest on 66 pairs.

Cohort	Set	$\widehat{\text{MMD}}^2$	$\times$ mean	$\times$ max	z_{cohort}	z_{perm}	$\hat{p}_{\text{perm}}$
17 subj. $K = 24$	null (276 pairs)	0.017 ± 0.005	—	—	—	—	—
	DDPM	0.095	5.5	2.3	15.7	423.3	< 0.002
	WDM-3D	0.225	13.0	5.5	42.0	709.2	< 0.002
	LIDC CT	0.410	23.6	10.1	79.3	719.1	< 0.002
	Retinal	0.475	27.4	11.7	92.4	810.2	< 0.002
50 subj. $K = 24$	null (276 pairs)	0.006 ± 0.002	—	—	—	—	—
	DDPM	0.081	14.5	6.9	49.5	381.6	< 0.002
	WDM-3D	0.216	38.8	18.5	138.7	689.4	< 0.002
	LIDC CT	0.402	72.2	34.4	261.0	711.1	< 0.002
	Retinal	0.461	82.7	39.4	299.4	810.4	< 0.002
100 subj. $K = 12$	null (66 pairs)	0.003 ± 0.001	—	—	—	—	—
	DDPM	0.079	30.3	15.9	89.0	347.2	< 0.002
	WDM-3D	0.220	84.3	44.1	253.3	631.4	< 0.002
	LIDC CT	0.403	154.4	80.8	466.0	678.3	< 0.002
	Retinal	0.463	177.2	92.7	535.6	766.2	< 0.002

generation. For each we recorded five properties answerable from the text: whether the size of the real reference set is stated; whether the number of distinct subjects behind it is stated; whether the selection procedure is described; whether train/test splits are stated to be subject-level; and whether the reference is drawn from the generator’s training data. Table 5 gives the result.

Three results stand out. *No paper in the sample states how many distinct subjects its real reference set is drawn from.* No paper fully states the reference set size, and none describes the selection procedure well enough to reproduce it. Only three of ten state that their splits are subject-level, and only one states explicitly whether the reference overlaps the generator’s training data — and that one does so because the overlap is a deliberate part of its design.

Generator distances against the real cohort-heterogeneity null

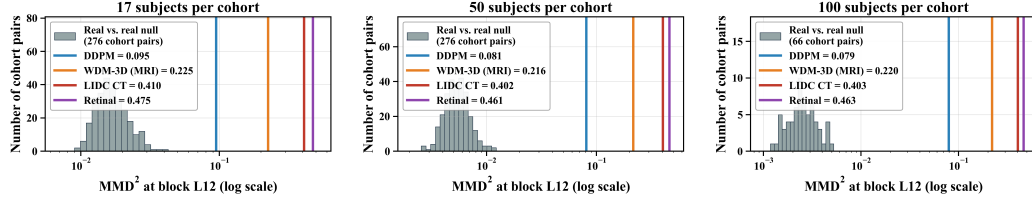


Figure 2: Generator distances against the real cohort-heterogeneity null at three cohort sizes. Histograms are pairwise distances between disjoint real patient blocks; vertical lines mark generators. The horizontal axis is logarithmic, because the null lies roughly two orders of magnitude below the generator distances and is not visible on a linear scale. As cohorts grow the null shifts left and concentrates, while the generator distances stay essentially fixed — the separation the evaluation can resolve is set by the reference, not by the generators.

The consequence is not that these papers are wrong. It is that a reader cannot distinguish a reference drawn from 17 subjects from one drawn from 400, and Section 4.1 shows those are different instruments giving answers that differ by a factor of two. The information needed to tell them apart is, in this sample, never present. That is what makes reference composition a reporting problem rather than a modelling one, and it is why our recommendations in Section 6 are about what to report rather than about what to compute.

Scope of the audit. This is a purposive sample of ten papers assembled from literature searches on medical image synthesis with distributional metrics, not a systematic review with a pre-registered protocol, and it is small enough that the percentages should be read as indicative rather than as prevalence estimates. Judgements are made from the main text; a detail buried in a supplement or a code release would be scored **X** here. We report the per-paper evidence for every judgement in the released audit record so that individual scores can be checked and disputed.

4.4. Encoder depth is task-dependent

All results in this section use the DISJOINT reference protocol of Section 3.2 at $N = 500$, so the cohort effects of Section 4.1 are controlled. Table 6 and Fig. 3 summarise every task at every block.

Table 5: Reference-set reporting in ten medical image generation papers. ✓ = stated, ~ = partially stated, ✗ = not stated, “?” = cannot be determined from the text. One paper uses no distributional metric and is marked — in the metric-specific columns. Percentages exclude that paper.

Paper	Metric	N imgs	N subj	Selection	Subj. split	Ref. from train
Pinaya et al. 2022	FID	✗	✗	✗	✗	?
Khader et al. 2023	MS-SSIM	—	—	—	✗	—
Akbar et al. 2023	FID	✗	~	✗	✓	?
Kong et al. 2023	FID	✗	✗	✗	✗	?
Woodland et al. 2024	Fréchet	~	✗	~	✗	?
Hashmi et al. 2024	FID	✗	✗	✗	✗	~
BrainMRDiff 2025	MMD	✗	✗	✗	✓	?
FADM 2024	FID	✗	~	✗	✓	?
Rajesh et al. 2025	FID, ECS	✗	✗	✗	✗	✓
Perceptual Eval. 2025	reader	~	✗	✗	✗	?
Fully stated		0/9	0/9	0/9	3/10	1/9

Fidelity peaks before the final block. Permutation z for the real-versus-DDPM comparison rises steeply through the early blocks, plateaus over L7–L9, and then declines. The maximum is at L8 ($z = 439.3$), with L7 (436.9) and L9 (436.6) statistically indistinguishable from it. The final block gives $z = 342.4$, which is 22.1% below the plateau. Reading the last layer, the default in work that uses a foundation model as a metric backbone, therefore discards roughly a fifth of the available discriminability on this comparison.

The raw discrepancies tell the same story: $\widehat{\text{MMD}}^2$ against the DDPM is 0.1042 at L7 and 0.0732 at L12, while the real-versus-real null is ≤ 0.0003 at every block. The decline into the final blocks is a loss of signal, not a change of scale.

Severity ordering, by contrast, is depth-robust. Every one of the twelve blocks recovers the a-priori ordering $\text{real} < \text{DDPM} < \text{WDM-3D} < \text{CT} < \text{retinal}$ with Spearman $\rho = +1.000$. If the only requirement is to rank generators of clearly different quality, depth does not matter; if the requirement is to resolve two similar generators, it does.

Near-duplicate detection improves with depth. Memorization recovery error falls essentially monotonically with depth (Table 6, second column): mean absolute error is 0.512 at L1, 0.027 at L6, 0.007 at L9 and 0.004 at L10–L11. Shallow blocks are not merely worse, they are unusable: at L1 the

detector recovers essentially none of the injected duplicates. The behaviour is stable across jitter scales up to $\sigma = 0.02$ and collapses for all depths at $\sigma \geq 0.1$, where the perturbation has destroyed the near-duplicate relationship the detector is meant to find.

Running the identical sweep under the HEAD protocol reverses this ordering: there, shallow blocks appear to outperform mid-depth ones. Both protocols are reported here from the same code and the same generated set, so the reversal can be checked directly rather than taken on trust. The HEAD result is an artefact of the 17-subject reference: when the reference spans few subjects, its within-reference nearest-neighbour distances are small, the detection threshold is correspondingly tight, and the ranking across depths inverts. We report this explicitly because it is a concrete instance of the paper’s general point — the reference protocol changed the sign of a methodological conclusion, not merely the value of a number.

Degradation monotonicity fails only at the shallowest blocks. Under a Gaussian noise ladder, $\widehat{\text{MMD}}^2$ increases monotonically with σ at every block from L3 onwards ($\rho = +1.000$). L1 ($\rho = +0.857$) and L2 ($\rho = +0.893$) are non-monotone: mild noise initially *decreases* the measured discrepancy, because at those depths the representation is dominated by low-level texture statistics that noise makes more similar between the two sets before it makes them different. A metric read at L1 would report that a slightly corrupted generator had improved.

4.5. *k*-NN recall fails its mode-drop check at every depth

The mode-drop validation asks a minimal question: if diversity is deliberately removed from the generated set, does measured coverage fall? For *k*-NN recall (Kynkäänniemi et al., 2019) in this feature space, the answer is no at every one of the twelve blocks. The Spearman correlation between drop fraction and recall is *positive* everywhere, ranging from +0.821 at L12 to +1.000 at five blocks (Table 6, third column). Discarding 80% of the generated samples *increases* measured recall — at L9, from 0.008 to 0.114.

The mechanism is visible in the raw values and is a known property of the estimator rather than a pathology of the encoder. Recall is the fraction of real points falling inside the union of *k*-NN balls around generated points, and each ball’s radius is that point’s own *k*-th nearest-neighbour distance *within the generated set*. Removing generated samples increases every surviving point’s *k*-NN radius, inflating each ball. Below some density the inflation

Table 6: Depth $\times$ task grid for RadioDINO-s16, $N = 500$, patient-disjoint reference. Best value per column in bold; the a-priori choice for each task is marked $\dagger$. No single depth wins every task, and the final block wins none.

Block	Fidelity	Memorization	Coverage	Per-image AUC	
	$z_{\text{perm}} \uparrow$	MAE $\downarrow$	$\rho(d, \text{recall}) \downarrow$	centroid	Mahalanobis
L1	171.6	0.512	+0.900	0.592	0.461
L2	212.6	0.163	+1.000	0.664	0.725
L3	185.0	0.180	+1.000	0.646	0.622
L4	221.1	0.114	+0.900 $\dagger$	0.654	0.771
L5	267.4	0.083	+1.000	0.628	0.858
L6	326.5	0.027	+0.900	0.674	0.921
L7	436.9	0.013	+1.000	0.744	0.928
L8	439.3	0.016	+0.900	0.796	0.949
L9	436.6	0.007 $\dagger$	+0.975	0.846	0.972
L10	404.6	0.004	+1.000	0.840	0.972
L11	368.6	0.004	+1.000	0.821	0.967
L12	342.4 $\dagger$	0.008	+0.821	0.810	0.962
Best block	L8	L10–L11	none	L9	L9–L10

of the balls outpaces the loss of centres, and measured coverage rises. In a 384-dimensional self-supervised embedding, where the generated set occupies a thin region and absolute recall is already very low (0.008–0.044 at $d = 0$ for blocks L7–L12), the estimator is in that regime from the start.

We therefore report no coverage number in this feature space and recommend that others do not either without first running this validation. We note that the diagnostic may behave correctly in embeddings where the generated set is denser relative to the real manifold; the claim here is specific to the setting, not a general refutation of the estimator.

4.6. Per-image scoring depends more on the distance than on the depth

Two per-image tasks were run: separating held-out real slices from cross-modality CT, and separating them from the DDPM’s own output. The first is saturated — every rule at every depth achieves $\text{AUC} \geq 0.997$ — and is therefore useless for comparing rules or depths. We note this because cross-modality OOD detection is a common headline experiment in this literature, and a saturated benchmark cannot support the conclusions usually drawn from it.

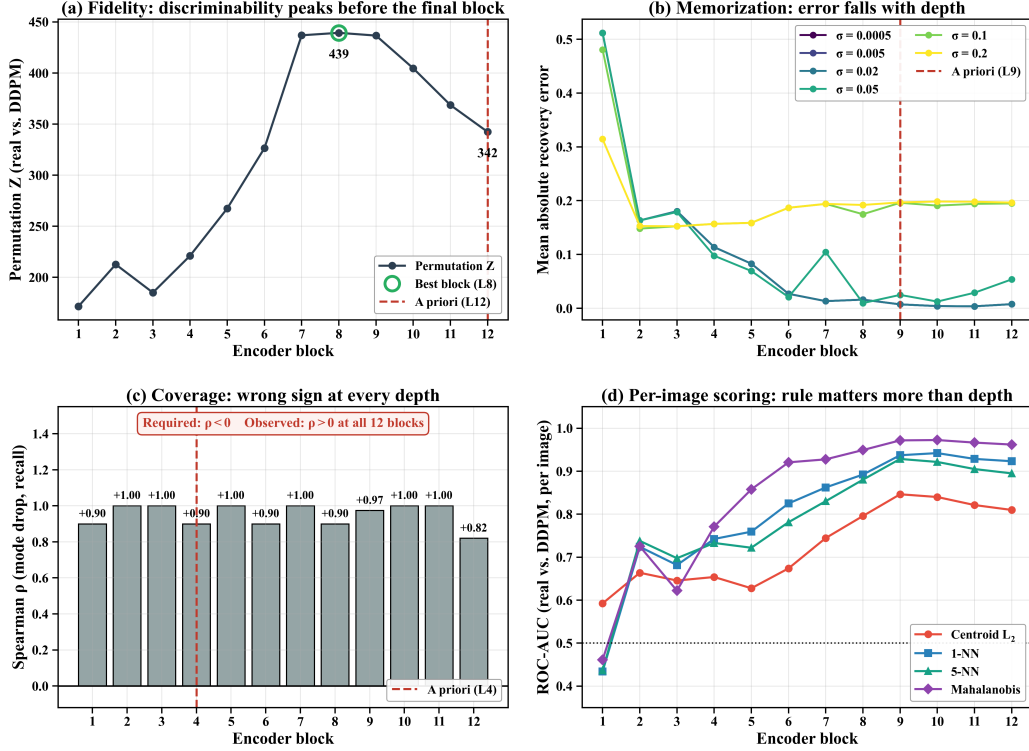


Figure 3: Depth profiles for four evaluation tasks. (a) Fidelity discriminability peaks at L7–L9, not at the final block. (b) Memorization recovery error falls with depth; curves are jitter scales, and the task collapses for all depths at large σ . (c) k -NN coverage: the Spearman correlation between mode drop and recall is positive at every depth, the opposite of the required sign. (d) Per-image scoring: the choice of distance separates the curves more than depth does. Dashed vertical lines mark the a-priori depth for each task.

On the discriminative same-modality task, the ranking of rules is consistent across depth and larger than the effect of depth itself (Table 6, last two columns). At the best depth, shrinkage Mahalanobis reaches AUC 0.972, 1-NN 0.942, 5-NN 0.929, and distance to the reference centroid 0.846. The centroid rule is worst at every block from L2 onward, by a margin of 9–23 AUC points. This is expected on geometric grounds: a centroid distance assumes an isotropic, unimodal reference, whereas a medical cohort is neither, and a shrinkage Mahalanobis distance accounts for the anisotropy that the raw Euclidean distance ignores. Where a per-image score is required — for flagging individual synthetic images before clinical use, for instance — the distance should be chosen deliberately rather than defaulted to a centroid.

4.7. Cross-metric comparison on a single protocol

Table 7 reports every metric on one reference protocol and one implementation per metric, which Section 4.1 argues is necessary for the numbers to be comparable at all. Two rows contain no generated images. REAL PROBE is a second, subject-disjoint block drawn by the same BALANCED procedure as the reference: the value a metric should return when nothing is wrong. REAL HEAD is the block that the default filename-ordered protocol selects — also entirely real BraTS slices, at the same resolution, from the same collection, preprocessed identically.

A real reference can look like a generative model. The two real rows are the practical summary of this paper. Drawn correctly, a real block sits at $\widehat{\text{MMD}}^2 = 0.002$ and $\text{FID} = 36.0$. Drawn by the convenient protocol, the same kind of comparison — real against real — reaches $\widehat{\text{MMD}}^2 = 0.067$ and $\text{FID} = 64.6$. The DDPM, an actual generative model, sits at 0.079 and 70.8. On FID the gap between a misdrawn real reference and a real generator is about six points, set against the 29-point spread between the two real rows. A reader given only the scalar 64.6 has no way to tell that no generative model was involved in producing it. This holds for every metric in the table, including the two natural-image baselines, so it is a property of the protocol rather than of the medical backbone.

Ordering, and where the backbones disagree. All four metrics order the three generated sets in the same way relative to the reference, and all place the weak same-modality generator closer than the cross-modality shift (Table 7). The metrics disagree on the *far* end of the ordering. The RadioDINO MMD places retinal fundus further from brain MRI (0.463 at L12) than chest CT (0.403); FID and KID place CT further (322.2) than retinal (248.4). Neither ordering is wrong in the abstract — they encode different notions of similarity. RadioDINO, trained on radiology, treats CT and MRI as neighbouring modalities and ophthalmic photography as remote; InceptionV3, trained on natural images, responds to surface appearance, and a colour fundus photograph is closer to its training distribution than a greyscale CT slice. For evaluating a CT or MRI generator we regard the radiological ordering as the more useful of the two, but this is an argument about which similarity a practitioner wants, not evidence that one backbone is more accurate.

We had previously reported, on a HEAD reference, that CMMD inverts the ordering of a weak same-modality generator against a cross-modality CT

Table 7: All metrics on one reference protocol (BALANCED(100), $N = 500$) and one implementation each. The first two rows contain only real images. Bootstrap intervals (60 resamples) are given for comparisons under the alternative; none is given for the real rows, where the statistic approaches degeneracy and the ordinary bootstrap is not consistent (Section 3.5).

Comparison set	Content	$\widehat{\text{MMD}}^2_{\text{L8}}$	$\widehat{\text{MMD}}^2_{\text{L12}}$	FID	KID
REAL PROBE	real vs. real	0.0016	0.0015	36.0	0.0009
REAL HEAD	real vs. real	0.0628	0.0674	64.6	0.0299
DDPM	real vs. gen.	0.1028	0.0792	70.8	0.0380
WDM-3D (MRI)	real vs. gen.	0.2155	0.2204	174.8	0.1574
LIDC CT	real vs. gen.	0.3886	0.4034	322.2	0.3725
Retinal	real vs. gen.	0.5026	0.4632	248.4	0.2851

shift. That measurement was made on the reference protocol this paper argues against, and we therefore do not carry the claim forward; whether it survives a subject-balanced reference is left open.

A note on the degenerate row. The bootstrap interval our procedure returns for REAL PROBE at L8 is $[0.0025, 0.0047]$ for a point estimate of 0.0016 — an interval that does not contain the estimate it is meant to bracket. This is not an implementation error but the expected behaviour of the ordinary bootstrap on a degenerate U -statistic (Arcones and Giné, 1992), and it is precisely why we characterise null behaviour with the empirical inter-cohort distribution of Section 3.5 rather than with bootstrap intervals. We report the pathology rather than suppressing it, because a reader who applies the bootstrap uncritically to a near-null comparison will meet it.

4.8. Lesion-level specificity

A metric intended for radiology should respond more strongly to changes in pathological tissue than to equivalent changes elsewhere. We test this on the LGG collection (Buda et al., 2019), which provides expert tumour masks. Gaussian noise of scale σ is applied either inside the annotated tumour region or inside an area-matched control region, and we report the ratio of the induced discrepancy in the two conditions. Two controls are used: a mirrored region at the contralateral position, and a texture-matched region selected to have comparable gradient energy (4.97×10^4 for tumour, 4.77×10^4 mirrored,

Table 8: Lesion selectivity: ratio of the discrepancy induced by perturbing tumour tissue to that induced by perturbing an area-matched control region. A value of 1.0 means no lesion selectivity. Bootstrap 95% intervals; † marks intervals excluding unity. Values at $\sigma \leq 30$ are omitted because the control response is near zero and the ratio is degenerate.

σ	RadioDINO MMD		Inception MMD	
	mirrored	texture-matched	mirrored	texture-matched
45	1.71 [0.94, 2.14]	1.81 [0.94, 2.12]	1.08 [0.92, 1.24]	1.03 [0.87, 1.16]
60	1.57 [1.03, 1.97] [†]	1.59 [1.02, 1.90] [†]	1.08 [0.93, 1.22]	1.05 [0.91, 1.17]
90	1.39 [1.06, 1.71] [†]	1.40 [1.03, 1.69] [†]	1.06 [0.93, 1.19]	1.06 [0.94, 1.17]

5.33×10^4 texture-matched), which controls for the possibility that the metric merely responds to local structural complexity.

The RadioDINO MMD is lesion-selective where an Inception-based MMD is not, but only in the upper part of the sweep (Table 8). At $\sigma = 60$ and $\sigma = 90$ the bootstrap interval excludes unity (1.57, [1.03, 1.97] and 1.39, [1.06, 1.71] against the mirrored control); at $\sigma = 45$ the point estimate is larger (1.71) but the interval [0.94, 2.14] includes unity. At $\sigma \leq 30$ the control-region response is near zero and the ratio is numerically degenerate, so we report no value. The Inception baseline stays within 1.01–1.08 throughout, with intervals covering unity at every σ .

We state this as suggestive rather than established, for two reasons. The effect is individually significant at only two of the six perturbation levels tested. More importantly, the perturbation is additive Gaussian noise confined to a mask, which is a proxy for pathology-relevant change rather than an instance of it: it establishes that the embedding weights tumour regions differently from matched healthy regions, but not that it tracks clinically meaningful pathology. Establishing the latter would require pathology-preserving edits, lesion insertion or removal, or agreement with radiologist judgement, none of which we attempt here.

5. Discussion

5.1. What a reported metric value does and does not mean

The results in Section 4.1 put a bound on how much a single reported number can carry. Two studies can both report “MMD² = 0.15 against 500 real BraTS slices” while differing by a factor of two in the quantity

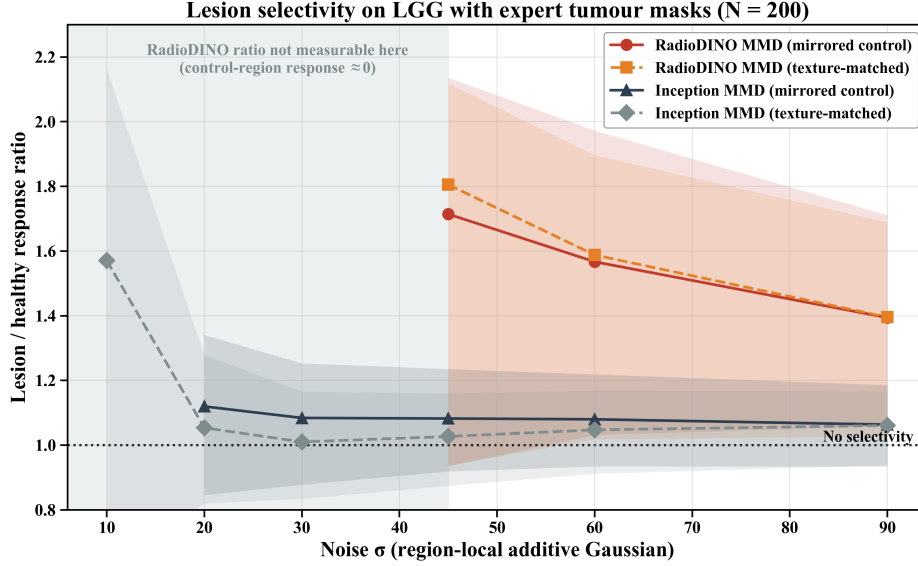


Figure 4: Lesion selectivity across the perturbation sweep on LGG, with area-matched mirrored and texture-matched controls. Shaded bands are bootstrap 95% intervals. The shaded region at low σ marks where the control-region response is near zero and the ratio is not measurable.

they intend to describe, purely through reference composition. Nothing in the number, the sample size, or the estimator’s unbiasedness reveals the difference. Unbiasedness is a property of the estimator with respect to a fixed pair of distributions; it says nothing about whether the empirical reference represents the population it stands for.

This is the generative-evaluation counterpart of a problem that discriminative medical imaging settled some time ago. Subject-level rather than image-level splits are standard there, because slices from one subject leak between train and test (Varoquaux and Cheplygina, 2022), and multi-site datasets are known to carry recoverable scanner signatures (Glocker et al., 2019). Generative evaluation inherited neither convention, because it inherited its protocols from natural-image benchmarks where neither problem exists. The reference set in a generative study plays the structural role that a test split plays in a discriminative one, and it deserves the same care.

5.2. Why the conventional significance test is the wrong instrument

Attaching a permutation test to a kernel two-sample statistic is technically correct and practically uninformative. Its null — that real and gener-

ated samples come from the same distribution — is false by construction for every generative model anyone would submit for evaluation. Our measurements bear this out: the test assigns $z > 340$ to an in-domain DDPM and to colour fundus photography alike (Table 4), separating neither from the other nor from anything else.

The behaviour under improving references is worse than merely uninformative. Because the permutation null is built by shuffling labels within the pooled sample, a more heterogeneous real set inflates the null variance and *depresses* the reported statistic. We say heterogeneous rather than representative deliberately: we have no sampling frame for the target population, so a wider subject draw is demonstrably more diverse but not thereby shown to be more representative. A practitioner who improves their reference from 17 to 100 subjects would see the conventional significance of their result fall, from $z = 423$ to $z = 347$, while the evaluation actually became better resolved. The inter-cohort statistic moves the other way, from 15.7 to 89.0.

We do not claim the inter-cohort reference distribution is the only correct construction. It requires subject labels, which not every released dataset provides; it is defined at a particular cohort size, so two studies must match cohort size to be comparable; and with K blocks it rests on $K(K - 1)/2$ non-independent pairs, so its tail should be read as descriptive rather than as a calibrated p -value. We report z_{cohort} and the ratio to the *mean* observed cohort distance for that reason, rather than leaning on p_{cohort} , whose resolution is bounded by the number of pairs, or on a ratio to the maximum, which is an order statistic that is not comparable across conditions contributing different numbers of pairs. What we do claim is that a null built from real cohorts answers the question a practitioner is actually asking, and that the permutation null does not.

5.3. Reading depth deliberately

The depth audit yields a practical result and a cautionary one. The practical result is that a single frozen encoder supports several evaluation tasks at different depths, at no additional cost: all twelve representations come from one forward pass, so reading L8 for fidelity and L10 for near-duplicate detection is free. The cautionary result is that the final block, which is what the literature reads by default, is the best choice for none of the four tasks we measured, and is 22% below the fidelity plateau. We name L8 as the maximum but not as “the” optimal depth: L7, L8 and L9 are not

distinguishable from one another, and the defensible statement is that the optimum lies in a mid-to-late plateau, not at the final block.

We are explicit that our depth recommendations are empirical rather than principled. We do not have a theory that predicts why fidelity discriminability should peak at L8 in this encoder, and we would not expect the specific number to transfer to a different backbone or a different modality. What we expect to transfer is the methodology: the depth profile is cheap to measure, the figures of merit above do not presume an answer, and any study using a foundation model as a metric backbone can run the audit for its own setting in minutes.

The memorization reversal described in Section 4.4 is worth restating as a general warning. On a HEAD reference we previously concluded that shallow blocks were better than mid-depth blocks for near-duplicate detection. On a subject-disjoint reference the ordering reverses and becomes monotone in the opposite direction. The reference protocol did not shift a number by a few percent; it inverted a methodological conclusion. Any depth study, ablation, or backbone comparison conducted on a convenience reference is exposed to the same risk.

5.4. Limitations

One dataset for the cohort analysis. The cohort-heterogeneity results are measured on BraTS, whose multi-site composition and subject-ordered filenames make the effect easy to isolate. We expect the phenomenon wherever a reference is assembled from a multi-site collection of volumetric studies, but we have not measured its magnitude elsewhere, and the specific inflation factors reported here should not be assumed to transfer. Whether datasets curated from a single site show a correspondingly smaller effect is an open question that the protocol in Section 3.5 is designed to answer.

One encoder for the depth analysis. The depth grid uses RadioDINO-s16. The finding that depth is task-dependent is likely general; the specific optima are not claimed to be.

Generators are not state of the art. The DDPM and WDM-3D samples serve as objects of measurement, not as contributions. A stronger generator would sit closer to the cohort null and would make the resolution argument of Section 4.2 more pressing, not less: the closer a generator comes to real data, the more the reference protocol determines whether the remaining gap can be detected at all.

Lesion-level specificity is a partial result. Using expert glioma masks with area-matched mirror and texture-matched controls, the RadioDINO MMD responds $1.4\text{--}1.8\times$ more strongly to perturbation confined to tumour tissue than to matched healthy tissue, where an Inception-based MMD responds at $1.05\text{--}1.08\times$. The effect is measurable only in the upper part of the noise sweep: bootstrap intervals exclude unity at $\sigma = 60$ (1.57, [1.03, 1.97]) and $\sigma = 90$ (1.39, [1.06, 1.71]) but not at $\sigma = 45$ (1.71, [0.94, 2.14]), and at $\sigma \leq 30$ the ratio is numerically degenerate because the control-region response is near zero. We report it as suggestive support for the domain backbone rather than as an established property.

Coverage. Our coverage result is a negative one about a specific estimator in a specific embedding. It does not establish that no coverage diagnostic can work here, and we did not evaluate alternatives such as α -precision/ β -recall (Alaa et al., 2022) at every depth. The recommendation is to run the mode-drop validation before reporting any coverage number, not to abandon coverage as a concept.

6. Recommended protocol

The following is what we would ask of a paper reporting a distributional metric for medical image generation, and what our released code implements by default.

1. **Report the number of distinct subjects behind every image set, not only the number of images.** A reference of 500 slices from 17 subjects and one from 400 are different instruments. This costs one number in a table and makes the remaining recommendations checkable.
2. **Build references by subject, not by filename.** Sample subjects first, then draw slices balanced across them and shuffled within them. Sorted-order convenience can select a structured subject subset with systematic feature differences.
3. **Keep real-versus-real comparisons subject-disjoint.** A null built by splitting slices within subjects measures near-duplicate distance and is optimistically small.
4. **Contextualise against inter-cohort variation.** Report z_{cohort} and the ratio to the *mean* real-versus-real cohort distance, at a stated cohort

size, alongside the raw score. Avoid ratios to the maximum: it is an order statistic whose expectation grows with the number of cohort pairs compared. Report the permutation p -value if desired, but do not read it as evidence of generator quality.

5. **Audit read-out depth; do not copy ours.** Run the depth audit for your own encoder — it costs one forward pass. Our optimum (L7–L9 for fidelity, with those three not distinguishable from one another) is a property of this encoder and this data, and should not be transplanted.
6. **Validate every diagnostic before reporting it.** Mode drop must reduce measured coverage; injected duplicates must be recovered at the injected rate. A diagnostic that fails its own validation should not appear in a results table.
7. **Use one implementation per baseline metric.** FID computed from two different Inception variants differs by tens of points on identical images.

7. Conclusion

We set out to audit a domain-pretrained embedding for generative medical image evaluation and found that the more consequential variable lies upstream of the embedding entirely. On BraTS, how the real reference cohort is assembled changes the reported discrepancy by up to a factor of two, moves the real-versus-real null by more than it moves the real-versus-generated score, and — in the worst case that the most convenient protocol happens to select — produces a distance between two sets of real images that exceeds the distance between real and generated ones. Reference composition is not a second-order detail in this setting; it is comparable in magnitude to the effect being measured.

We proposed calibrating against a null built from distances between disjoint real patient cohorts, which asks whether a generator differs from the reference by more than routine inter-cohort variation. Unlike the conventional permutation test, which assigns overwhelming significance to every generator and becomes *less* significant as the reference improves, the cohort-calibrated statistic tracks reference quality in the right direction and yields an interpretable effect size.

Within that protocol, a depth-resolved audit of RadioDINO-s16 shows that encoder depth is task-dependent and that the final block — the default

read-out — is optimal for none of the four tasks measured, falling 22% below the fidelity plateau at L7–L9, whose members are not distinguishable from one another. One widely used diagnostic, k -NN precision/recall coverage, fails its own mode-drop validation at every depth through neighbourhood-radius inflation, and we recommend against reporting it in this feature space without validation. For per-image scoring, the choice of distance matters more than depth.

None of these findings requires a new metric. They ask for a more careful account of what an existing one is measured against.

Code and data availability

Code implementing the metric, the inter-cohort reference distribution, the depth audit, the literature audit record and every experiment reported here is available at <https://github.com/Nishan-Charlie/m3-cohort-audit>. BraTS 2021 (Baid et al., 2021) and the LGG collection (Buda et al., 2019) are available from their respective repositories under their own terms; we redistribute no patient data. Scripts to regenerate every table and figure from raw data are included, together with the exact subject lists defining each cohort block.

CRedit authorship contribution statement

Sathiyamohan Nishankar: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review & editing, Visualization.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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